Power electronics Project Part C

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Question 1 – Theoretical

1.

a) Using KVL we can write out

$$V_m \sin(\omega t) = Ri(t) + L\frac{di(t)}{dt}$$

Expressing current as sum $i(t) = i_f(t) + i_n(t)$

We can get $i_f(t)$ from phasor analysis where

$$Z = \sqrt{R^2 + (\omega L)^2} \quad \text{and} \quad \theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$$ we can write $i_f(t)$ as

forced response $i_f(t) = \frac{V_m}{Z}\sin(\omega t - \theta)$

natural response $i_n(t) = Ae^{-t/\tau}$ * τ is time constant $\tau = \frac{L}{R}$

Then able to write $i(t)$ as

$$i(t) = \frac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-t/\tau}$$

We can evaluate A by using intial condition for current

$$i(0) = \frac{V_m}{Z}\sin(0-\theta) + Ae^0 = 0 \Rightarrow A = -\frac{V_m}{Z}\sin(-\theta) = \frac{V_m \sin\theta}{Z}$$

Can then write $i(t)$ as

$$i(t) = \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-t/\tau}\right]$$

we can rewrite in terms of ωt

$$i(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-\frac{\omega t}{\omega \tau}}\right]$$

Substituting $\omega t = \beta \Rightarrow i(t) = \frac{V_m}{Z}\left[\sin(\beta - \theta) + \sin(\theta)e^{-\beta/\omega\tau}\right] = 0$

for

$$i(\omega t) = \begin{cases} \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-\frac{\omega t}{\omega\tau}}\right] & \text{for } 0 \le \omega t \le \beta \\ 0 & \text{for } \beta \le \omega t \le 2\pi \end{cases}$$

Figure 1derivation for general expression for current in half wave rectifier

$w = 2\pi f = 2\pi 50 = 100\pi \qquad \tau = \frac{L}{R} = \frac{7\times10^{-3}}{5}$

$w = 100\pi$

$\tau = 1{,}4\times10^{-3}s$

$\theta = Tan^{-1}\left(\frac{wL}{R}\right)$

$\theta = Tan^{-1}\left(100\pi \times 1{,}4\times10^{-3}\right)$

$V_m = V_{rms}\sqrt{2}$

$V_m = 15\sqrt{2} = 21{,}21V$

$\theta = 23{,}74 = 0{,}414\,rad$

$$Z = \sqrt{R^2+(wL)^2} = \sqrt{5^2 + (100\pi \times 7\times10^{-3})^2}$$

$Z = 5{,}462$

$Z = 5{,}46\,\Omega$

want to thus solve for when

$\sin(\beta-\theta) = 0$

where

$\beta-\theta = n\pi$ we will consider first cross $n=1$

$\beta-\theta = \pi \Rightarrow \beta = \pi+\theta$ we already found θ

so β is

$\beta = Tan^{-1}(100\pi\times1{,}4\times10^{-3}) + \pi = 3{,}56\,rad$

$$i(wt) = \begin{cases} \frac{V_m}{Z}\left[\sin(wt-\theta)+\sin(\theta)e^{-\frac{wt}{w\tau}}\right] & \text{for } 0 \le wt \le 3{,}56 \\ 0 & \text{for } 3{,}56 \le wt \le 2\pi \end{cases}$$

The final expression can be written as 3.56rad = 203.74

$$\boldsymbol{i}(\omega \boldsymbol{t}) = \frac{\mathbf{21.21}}{\mathbf{5.46}}\left[\mathbf{\sin}(\boldsymbol{\omega t} - \mathbf{0.414}) + \mathbf{\sin}(\mathbf{0.414})\,\boldsymbol{e}^{-\frac{\boldsymbol{\omega t}}{\mathbf{0.44}}}\right] \boldsymbol{for}\ \mathbf{0} \le \omega \boldsymbol{t} \le \mathbf{3.56}$$

$$\mathbf{0}\ \boldsymbol{for}\ \mathbf{3.56} \le \omega \boldsymbol{t} \le \mathbf{2}\pi$$

b. For half wave rectifier we would define current as follows

$$I_0 = \frac{1}{2\pi}\int_0^{\beta} i(\omega t)d(\omega)$$

$$I_0 = \frac{1}{2\pi}\int_0^{3.56} \frac{\mathbf{21.21}}{\mathbf{5.46}}\left[\mathbf{\sin(\omega t - 0.414) + \sin(0.414)}\, e^{-\frac{\omega t}{0.44}}\right]\boldsymbol{d(\omega t)}$$

$$\boldsymbol{I_0 = 1.29A}$$

c. We can then calculate RMS current as follows

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{\beta} i^2(\omega t)d(\omega)}$$

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{3.56} \left\{\frac{\mathbf{21.21}}{\mathbf{5.46}}\left[\mathbf{\sin(\omega t - 0.414) + \sin(0.414)}\, e^{-\frac{\omega t}{0.44}}\right]\boldsymbol{d(\omega t)}\right\}^2 d(\omega t)}$$

$$\boldsymbol{I}_{rms} = 1.98A$$

d. The power absorbed by the RL load is

$$P = I_{rms}^2 R$$

$$P = 1.98^2 \times 5 = 19.6W$$

e. The power factor is calculated as follows

$$PF = \frac{P}{V_{rms} I_{rms}}$$

$$PF = \frac{19.6}{15 \times 1.98} = 0.66\ lagging$$

Question 2 – Simulation in MATLAB

a. For AC voltage sources with a peak of approximately 21.21 from the 15V RMS and frequency of 50Hz. I attain the following waveform

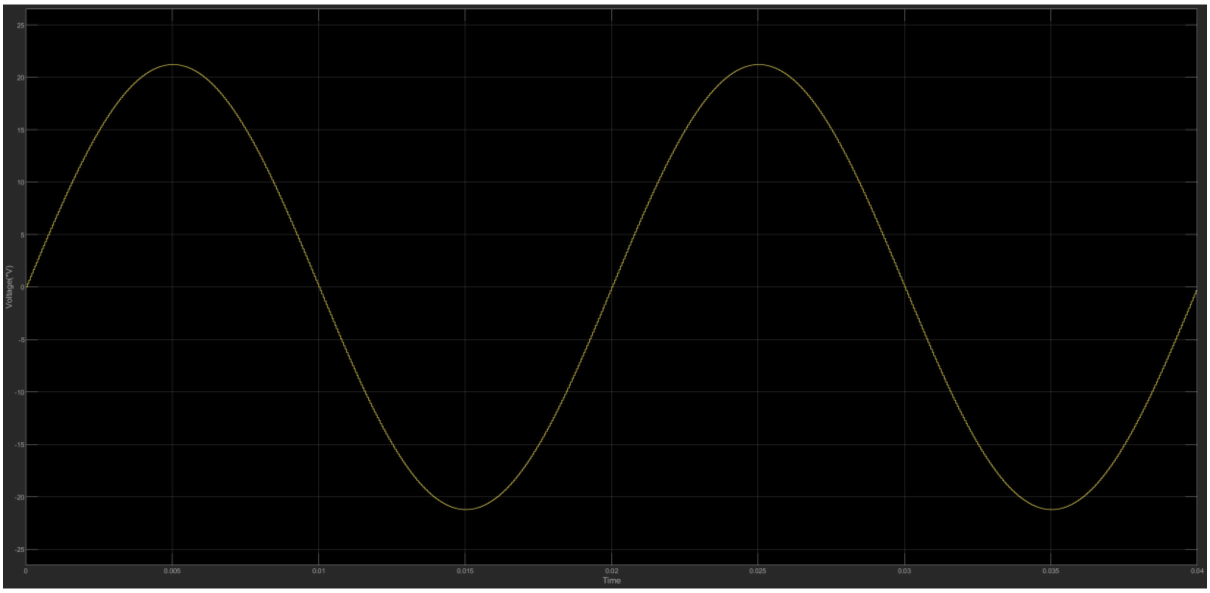


Figure 2 Source voltage for 2 cycles

b.

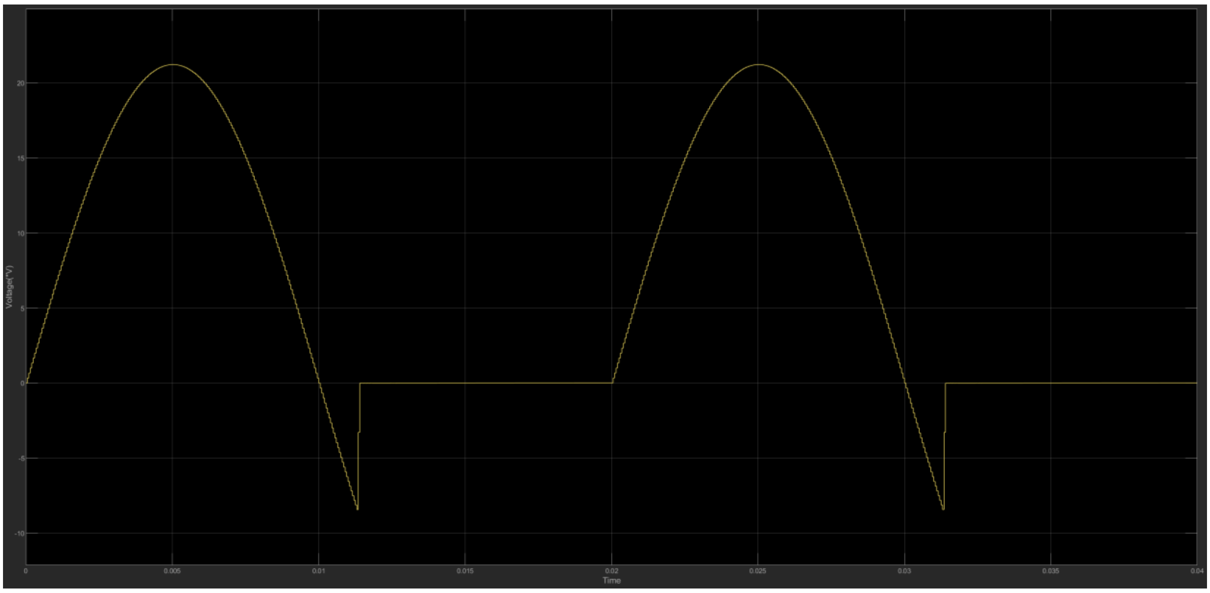


Figure 3 Load voltage

c.

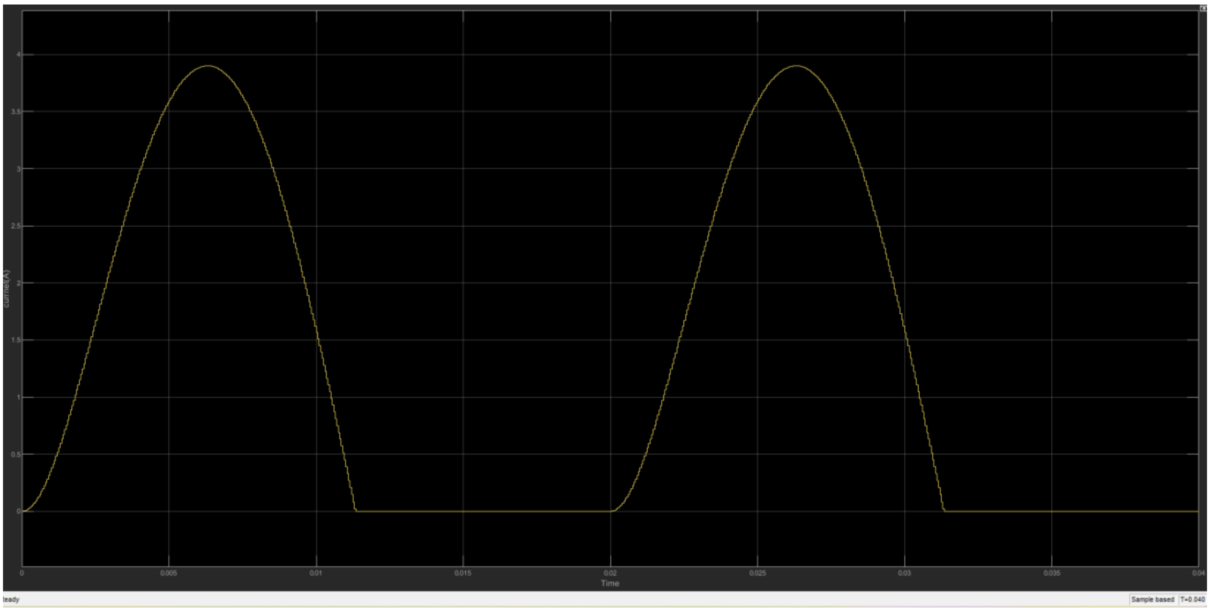


Figure 4 load current

d.

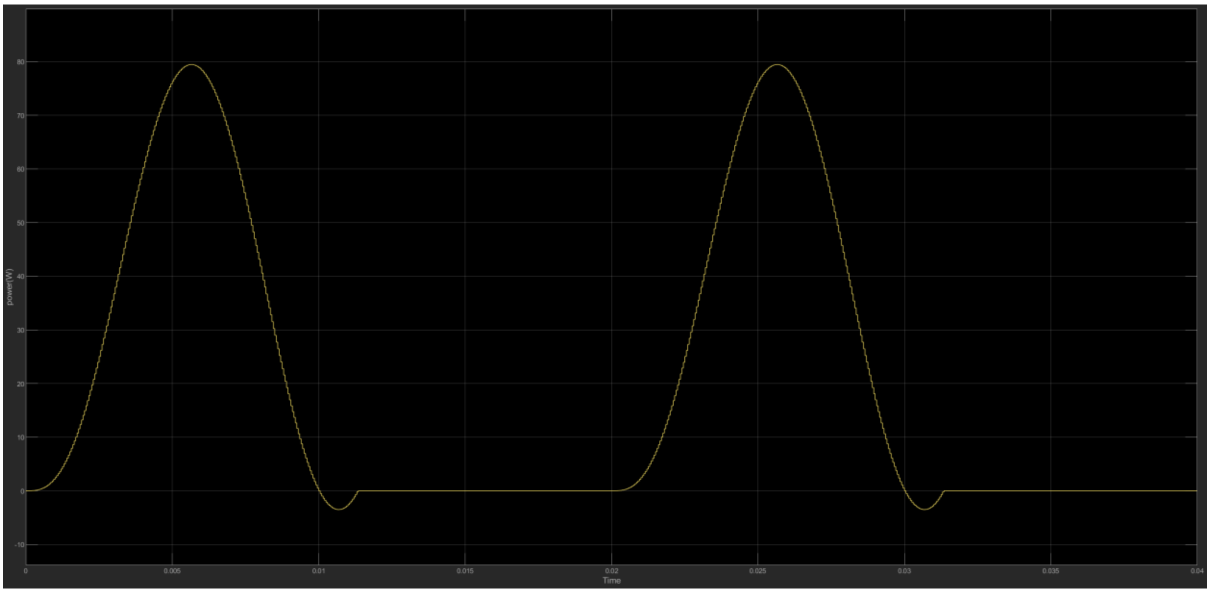


Figure 5 Instantaneous output power

2.

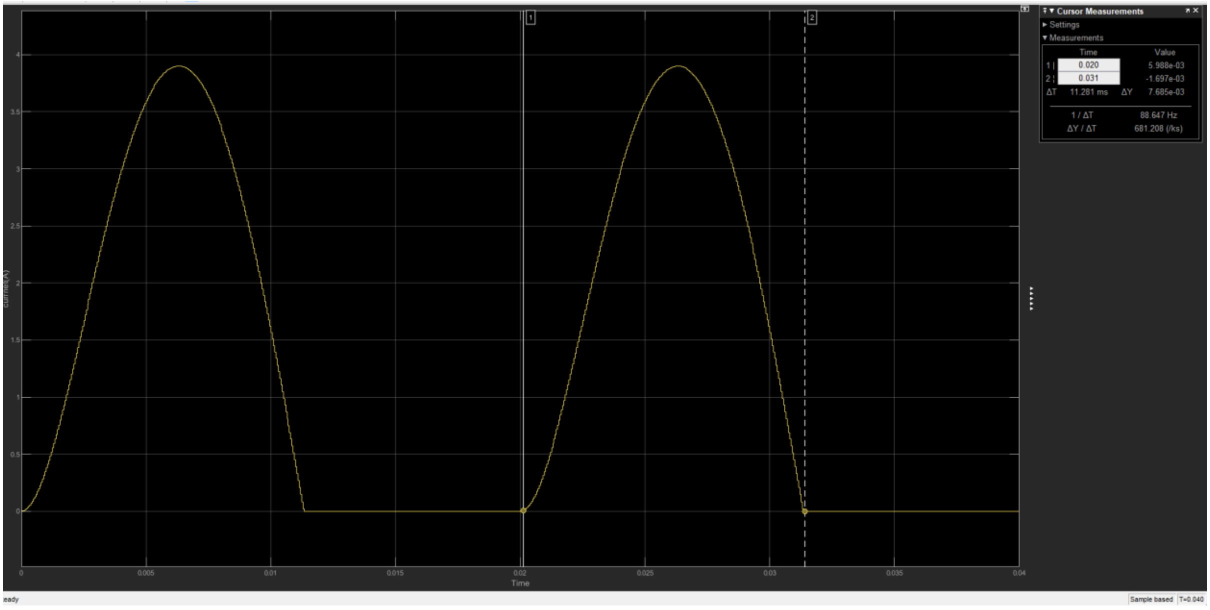


Figure 6 :Current waveform measurement for extinction angle

$$\beta = \frac{11}{20} \times 360 = 198\, degree \approx 3.46\, rad$$

We see that the extinction angle is lower than expected meaning the diode stops earlier than expected and so the energy is the inductor is dissipated faster causing load current to decay sooner

b.

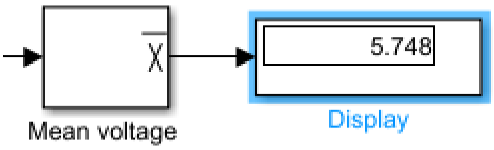


Figure 7 average voltage

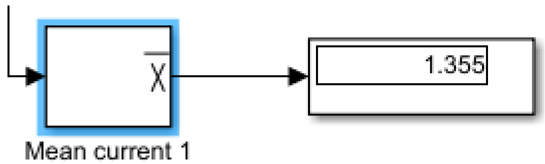


Figure 8 Average current

have that the measured value of the average current is 1.355A which is very close to the calculated 1.29A. which can be accounted for by consider that in most calculations we assume ideal passive components while Simulink may have parasitic resistances and simulate more like real world condition.

c.

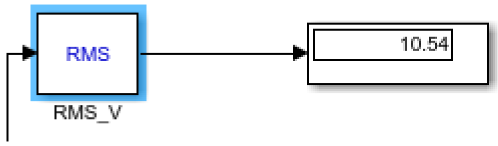


Figure 9 RMS load voltage

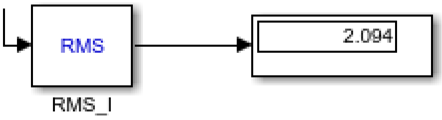
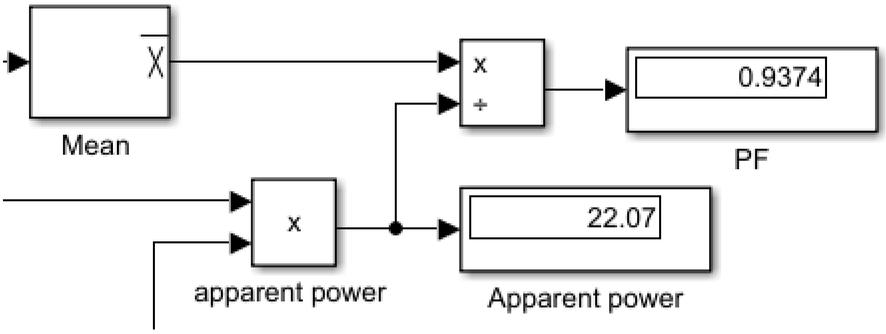


Figure 10 RMS current

The measured RMS values were 10.54V and 2.094A while the calculated values were 1.98A for RMS current which is very close to the measured value. Again, which can be accounted for the assumption of ideal components in calculation while as Simulink's acts to be more realistic.

d.



even with a simulation drop of 0V n=value swill still not match. The theoretical formulae make us of continuous calculus however in Simulink new are sampling at discrete intervals giving a sort of rounding effect but the PF value is still quite large of 0.9374. the displacement PF of the load itself is 0.915 which the measured PF is close to.

e.

Harmonic	Magnitude	Phase
0	6.46	90
1	10.78	2.58
2	5.00	-94.8
3	0.5657	40.23
4	1.237	-115.4
5	0.5217	6.594

3. This is due to stored energy in the inductive aspect of the circuit as the AC signal goes to zero the inductor releases its stored energy this causes D1 to conduct briefly even though the input is technically negative. This induced voltage keeps the diode forward biased forcing current to flow
4.

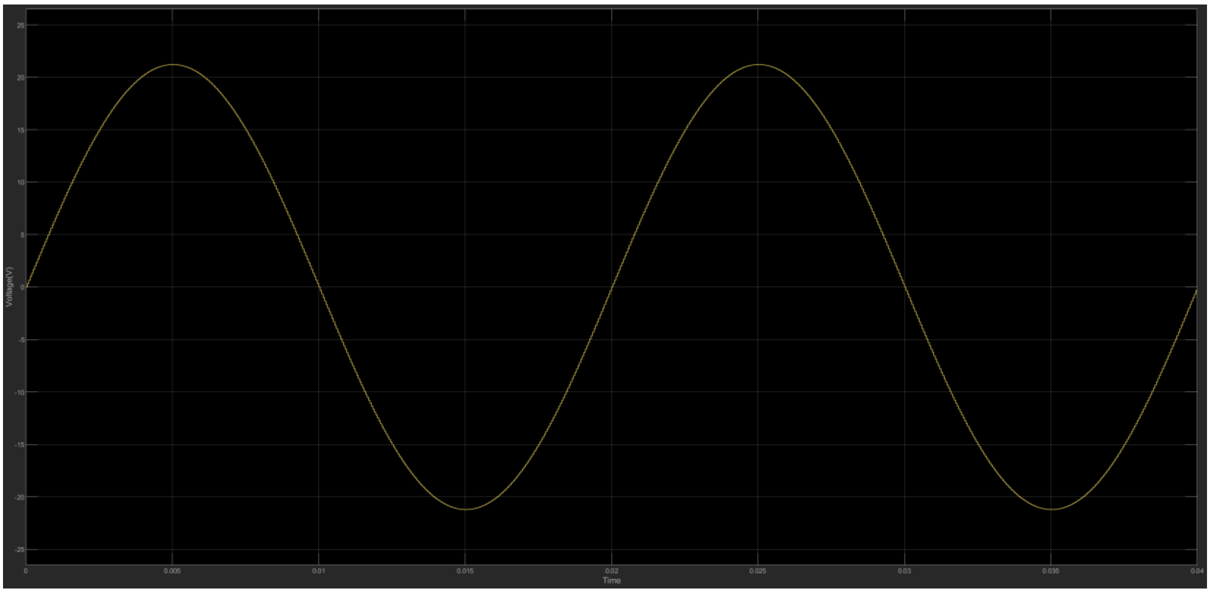


Figure 11 source voltage after adding free wheeling diode

As expected even with the addition of the freewheeling diode the voltage source remains unchanged.

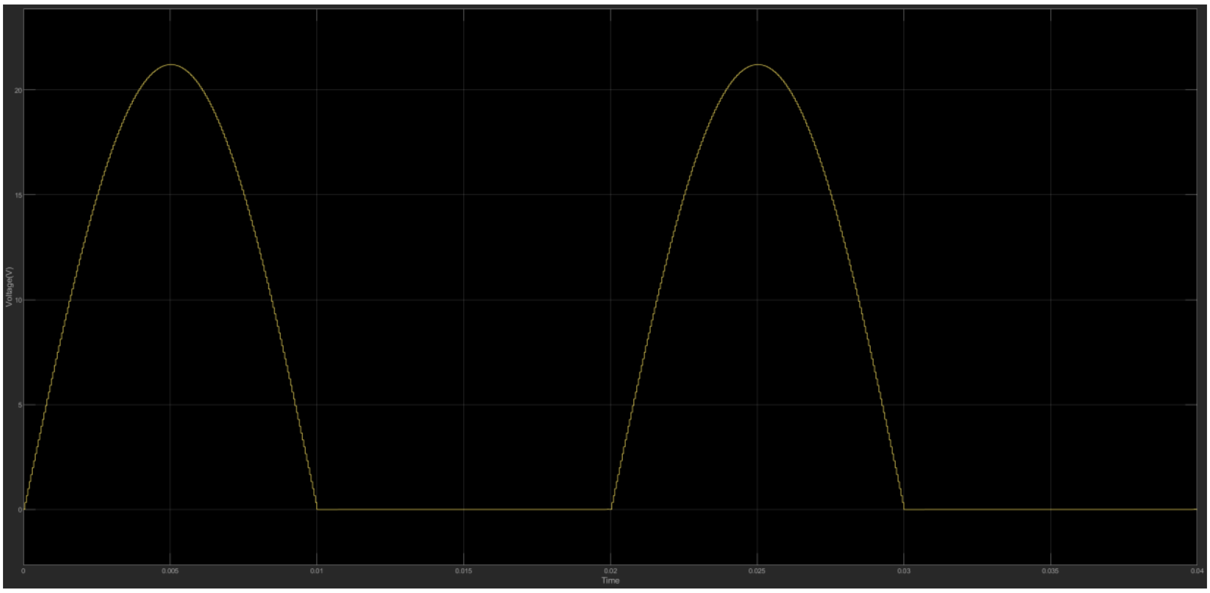


Figure 12 Load voltage with wheeling diode added across RL

We see the load voltage is now purely made up of the positive half cycle.

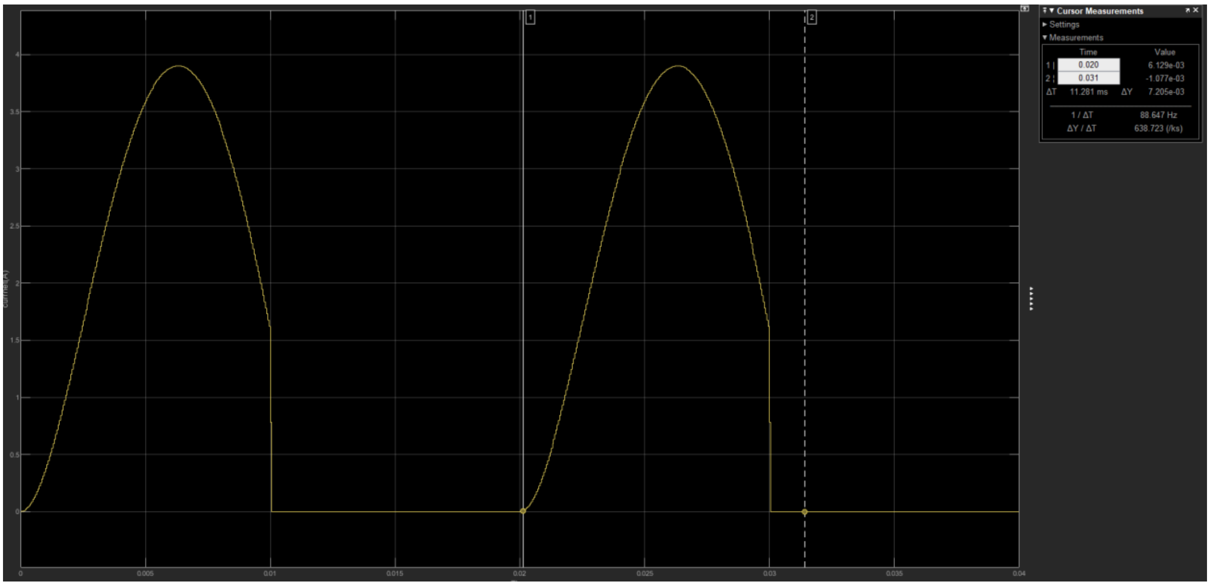
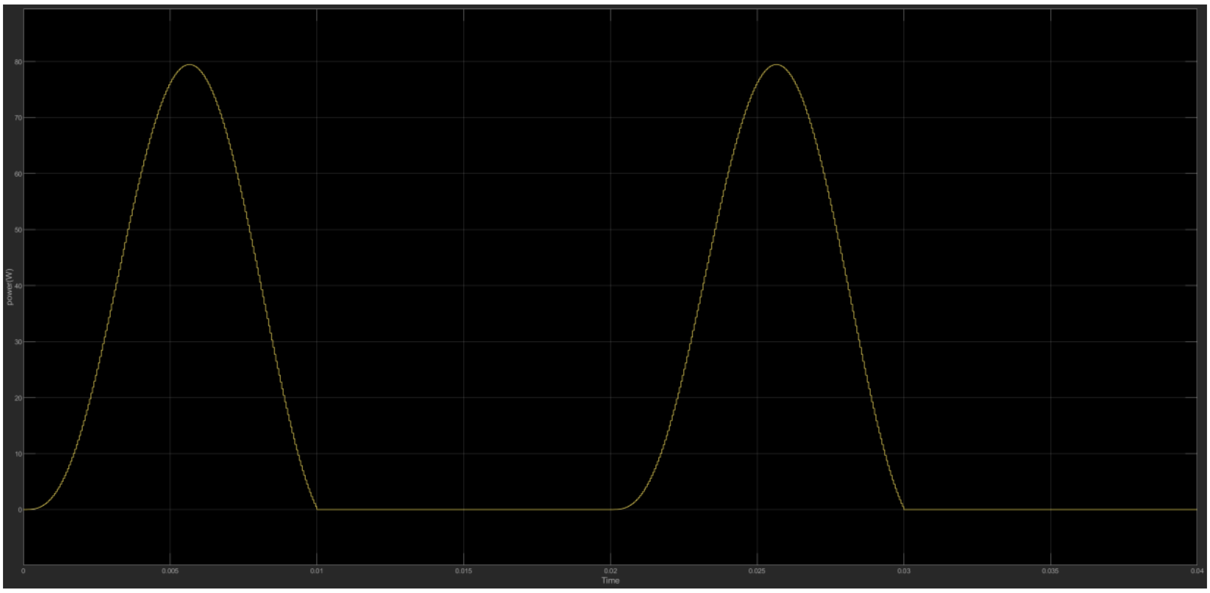


Figure 13 Load current after adding the wheeling diode across RL with cursor measurements

We see the load current is clipped.

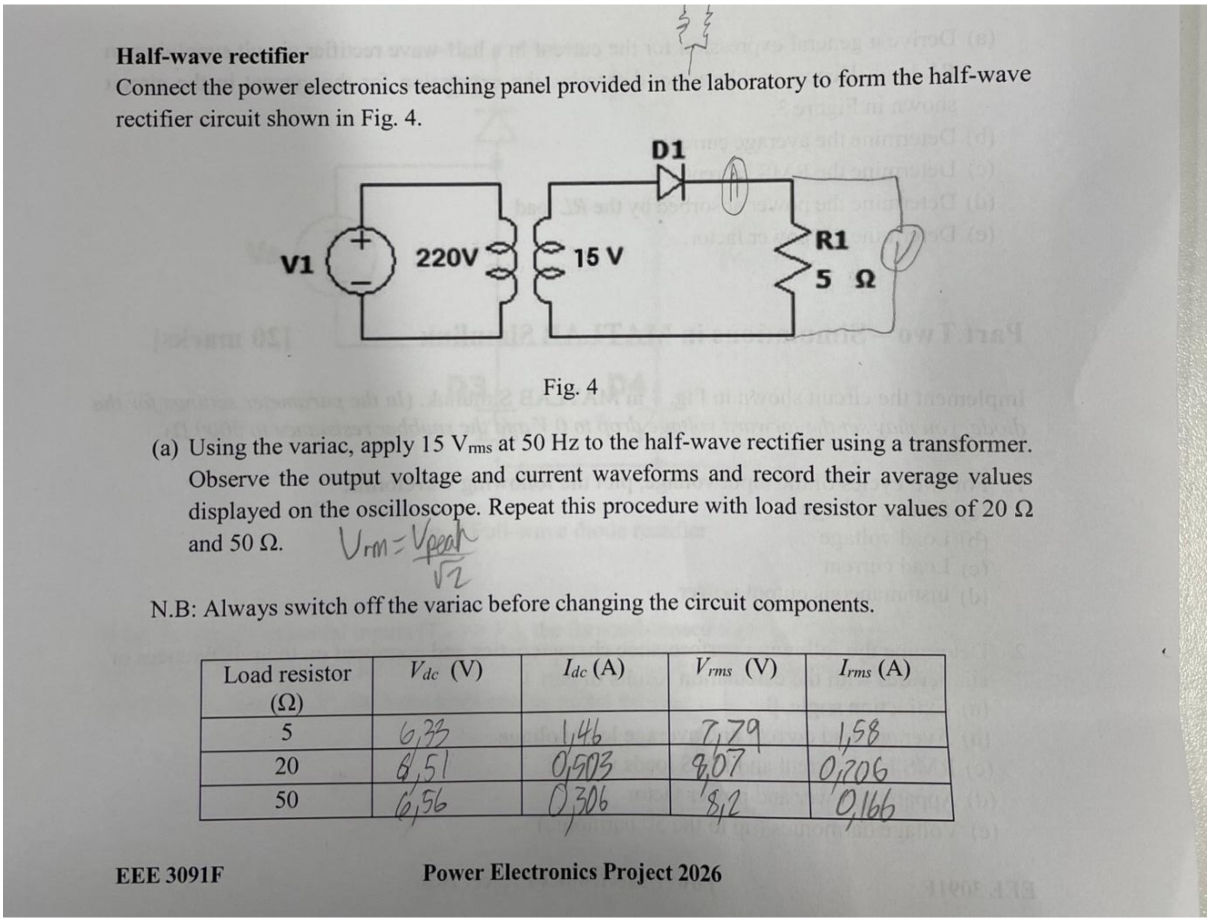


See that the voltage source waveform remains the same however now the load current and voltage have a smoother waveform rather than abruptly going to zero and also remain positive. And therefore, now the instantaneous output is always positive as well. These changes improve energy utilization and reduce voltage spikes across the load

5. Would need to store energy during the conducting half of the Ac cycle and then release it during the non-conducting half so as to keep the current flowing through the load without interruption. We can also opt to make the load time current big enough

Question 3 – Experimental results

a.



Half-wave rectifier

Connect the power electronics teaching panel provided in the laboratory to form the half-wave rectifier circuit shown in Fig. 4.

Fig. 4

(a) Using the variac, apply 15 V_{rms} at 50 Hz to the half-wave rectifier using a transformer. Observe the output voltage and current waveforms and record their average values displayed on the oscilloscope. Repeat this procedure with load resistor values of 20 Ω and 50 Ω. $V_{rm} = \frac{V_{peak}}{\sqrt{2}}$

N.B: Always switch off the variac before changing the circuit components.

Load resistor (Ω)	V_{dc} (V)	I_{dc} (A)	V_{rms} (V)	I_{rms} (A)
5	6,33	1,46	7,79	1,58
20	6,51	0,503	8,07	0,206
50	6,56	0,306	8,2	0,166

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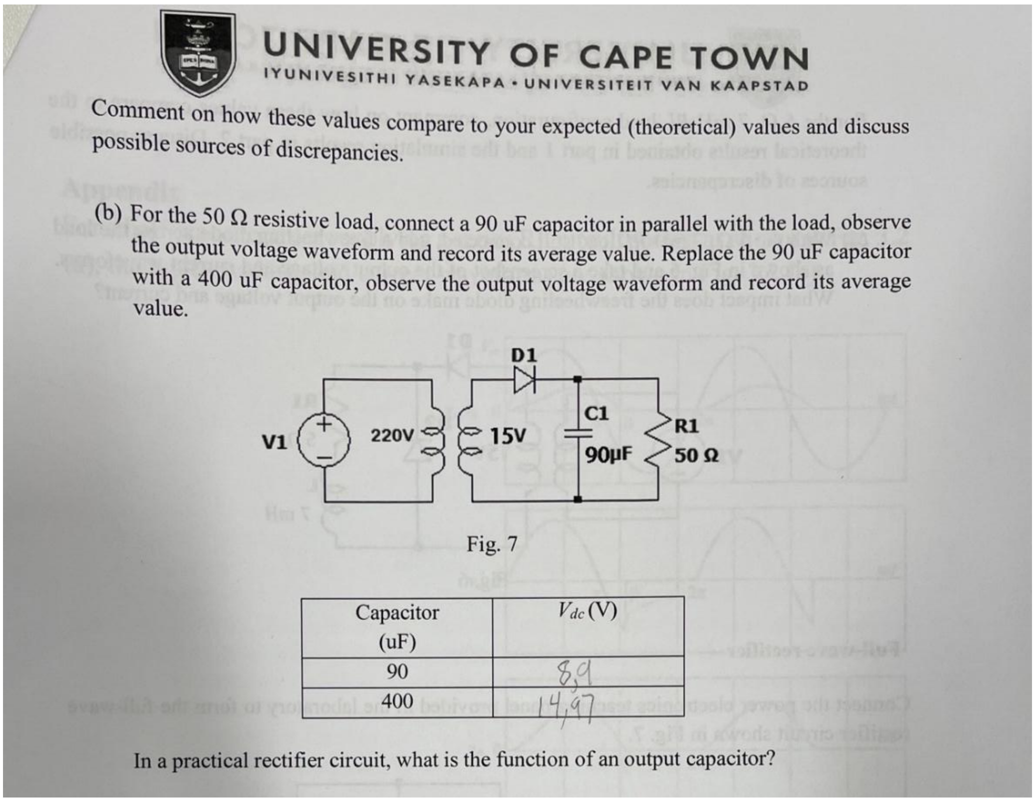
Figure 14 Half wave rectifier results across various loads

The calculated value for VDC closely resemble those obtained during the practical session. However, there was still some discrepancies that were noticed in the values for the RM voltage current and IDC. This can be due to the fact that in calculation we assume ideal values for components such as resistance or drops across diodes etc and are unable to account for tolerance in components values and inaccuracy in machine readings.

b.



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Comment on how these values compare to your expected (theoretical) values and discuss possible sources of discrepancies.

(b) For the 50 Ω resistive load, connect a 90 uF capacitor in parallel with the load, observe the output voltage waveform and record its average value. Replace the 90 uF capacitor with a 400 uF capacitor, observe the output voltage waveform and record its average value.

Fig. 7

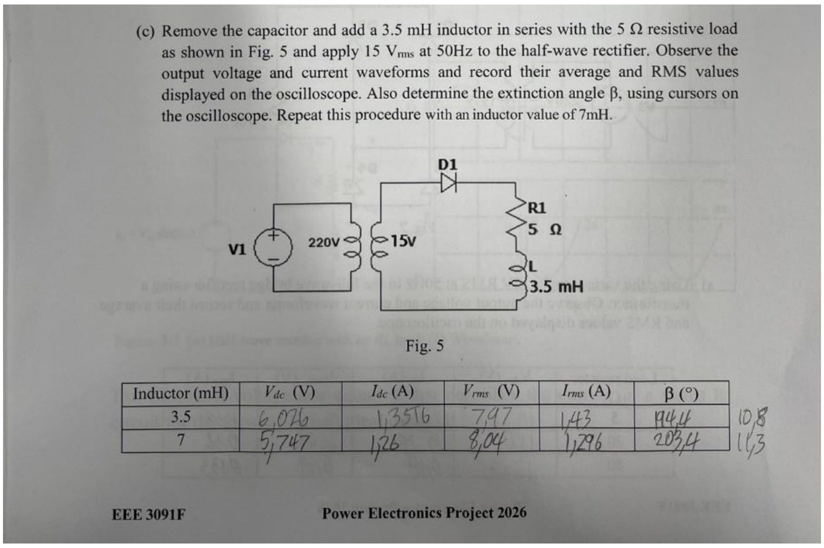
Capacitor (uF)	V_{dc} (V)
90	8,9
400	14,97

In a practical rectifier circuit, what is the function of an output capacitor?

Figure 15 half wave with an added capacitor in parallel

Capacitor can serve to function as a sort of filter and give a smoother DC voltage]

c.



(c) Remove the capacitor and add a 3.5 mH inductor in series with the 5 Ω resistive load as shown in Fig. 5 and apply 15 V_{rms} at 50Hz to the half-wave rectifier. Observe the output voltage and current waveforms and record their average and RMS values displayed on the oscilloscope. Also determine the extinction angle β, using cursors on the oscilloscope. Repeat this procedure with an inductor value of 7mH.

Fig. 5

Inductor (mH)	V_{dc} (V)	I_{dc} (A)	V_{rms} (V)	I_{rms} (A)	β (°)	
3.5	6,026	1,3516	7,97	1,43	194,4	10,8
7	5,747	1,26	8,04	1,296	203,4	11,3

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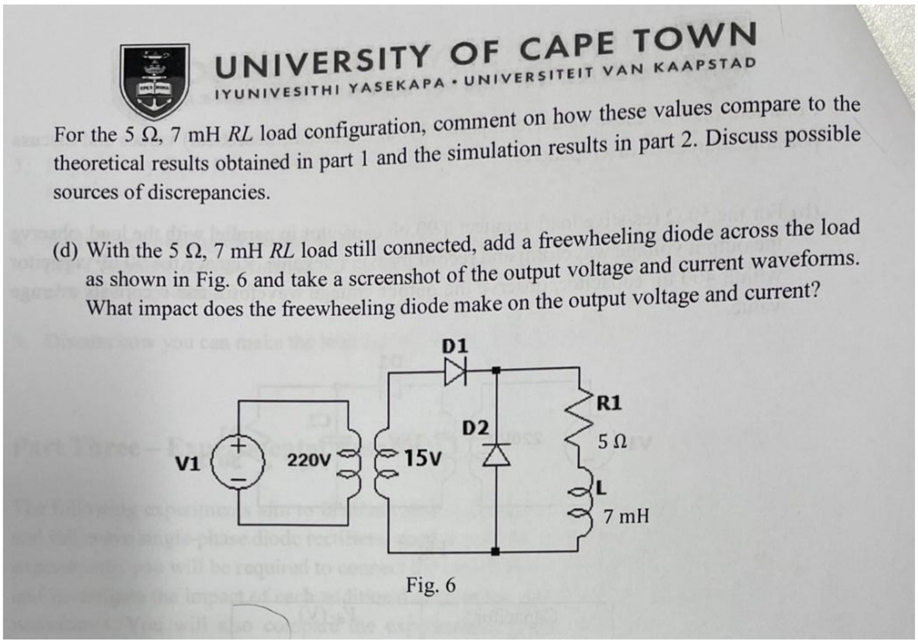
Figure 16 remove the capacitor and add inductor instead

Although the values found in the lab closely resemble those seen in the calculations done for part one there is still however noticeable discrepancies in the values which can most likely be attribute to the fact that in calculation, we assume ideal condition for components such as the inductor and resistor and as well for the source. While in actuality that is not the case and there are slight differences fluctuations present when measurement and selecting components which can account for the difference in values seen in the lab and calculation. Also due to the fact that we may not use the exact same point in calculation as we did in the lab and they can be off by small amount.

d.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For the 5 Ω, 7 mH *RL* load configuration, comment on how these values compare to the theoretical results obtained in part 1 and the simulation results in part 2. Discuss possible sources of discrepancies.

(d) With the 5 Ω, 7 mH *RL* load still connected, add a freewheeling diode across the load as shown in Fig. 6 and take a screenshot of the output voltage and current waveforms. What impact does the freewheeling diode make on the output voltage and current?

Fig. 6

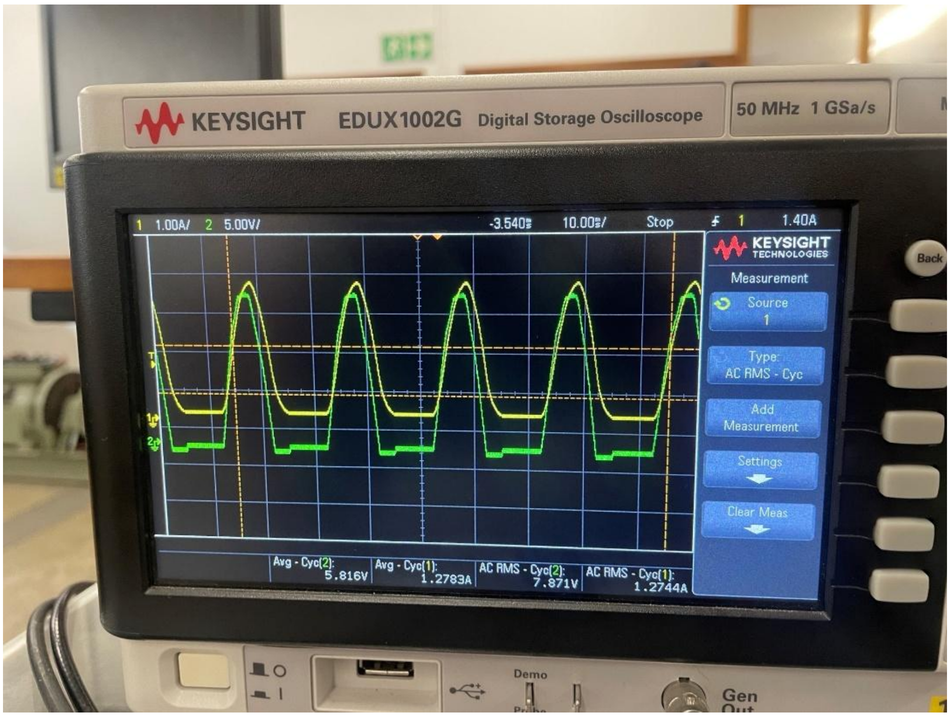


Figure 17 Screenshot of output voltage and currents waveforms

With the addition of the freewheeling diode, we are able to see that the diode has an impact on the waveforms for both the load voltage and current and acts to removes the dip as previously seen in the previous waveforms for voltage and current and produces a smoother curve as well as no spike in values.

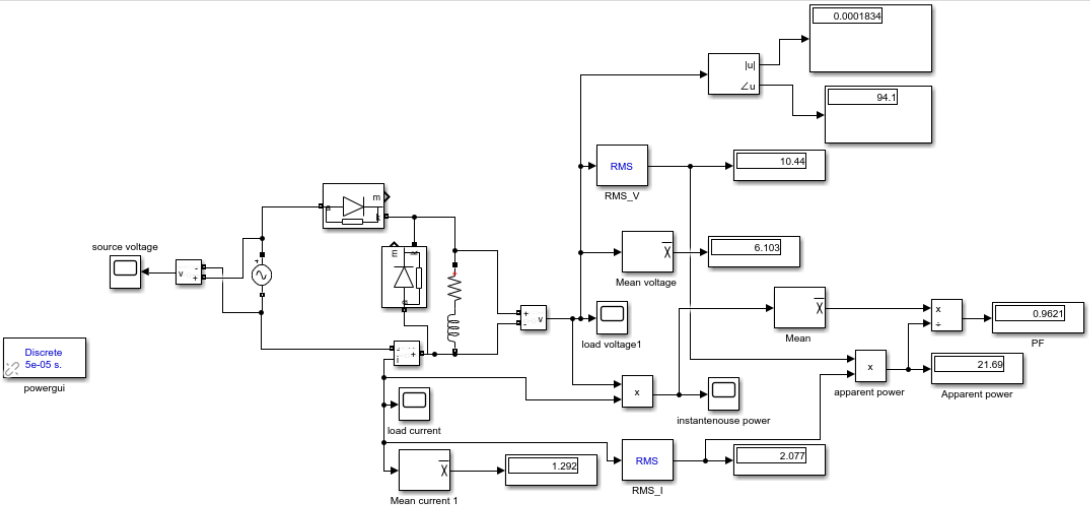
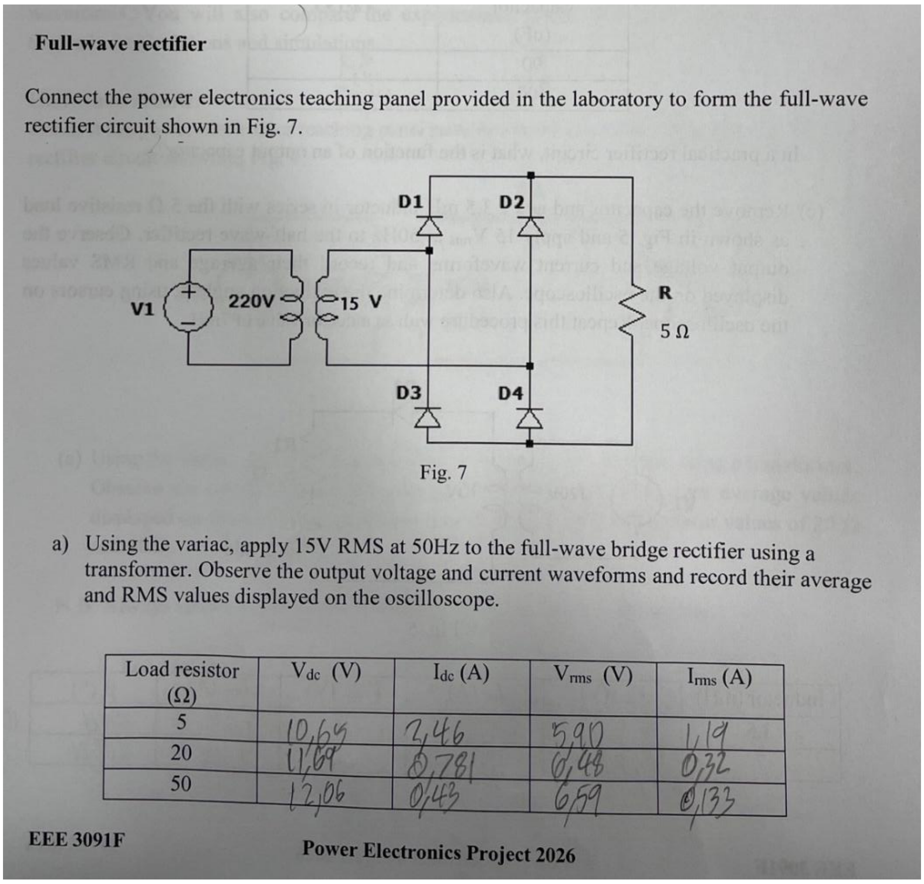


Figure 18 MATLAB circuit

FULL WAVE RECTIFIER

a.



Full-wave rectifier

Connect the power electronics teaching panel provided in the laboratory to form the full-wave rectifier circuit shown in Fig. 7.

Fig. 7

a) Using the variac, apply 15V RMS at 50Hz to the full-wave bridge rectifier using a transformer. Observe the output voltage and current waveforms and record their average and RMS values displayed on the oscilloscope.

Load resistor (Ω)	V_{dc} (V)	I_{dc} (A)	V_{rms} (V)	I_{rms} (A)
5	10,64	2,46	5,90	1,19
20	11,69	0,781	6,48	0,32
50	12,06	0,43	6,59	0,133

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